

Short-Term Deterioration of Visual Acuity and Visual Fields in Nonarteritic Anterior Ischemic Optic Neuropathy



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- **PURPOSE:** To determine the frequency and severity of further visual loss experienced by patients within ten weeks from diagnosis of acute nonarteritic anterior ischemic optic neuropathy (NAION).
- **DESIGN:** Retrospective case series.
- **METHODS:** Electronic medical records (EMR) at an academic neuro-ophthalmology practice were searched for diagnosis of “NAION” and all identified charts were reviewed to determine eligibility. Patients diagnosed with acute NAION between February 2014 and December 2023 who presented within four weeks of symptom onset and were seen in follow-up within ten weeks were included. Clinically significant decline in best corrected visual acuity (BCVA) and peripheral VF were defined as decline of BCVA ≥ 2 Snellen lines and decrease of ≥ 2 decibels (dB) in mean deviation (MD) on perimetric testing.
- **RESULTS:** Forty-nine eyes met inclusion and exclusion criteria. Sixty-seven percent of patients were male and average age at presentation was 66 years. Twenty-two percent of eyes demonstrated worsening of BCVA by ≥ 2 lines. Of these, 55% worsened by ≥ 4 lines and 27% by ≥ 8 lines. In 27% of eyes MD on perimetry worsened by ≥ 2 dB and in 18% by ≥ 4 dB. In total, 41% of eyes demonstrated clinically significant worsening of BCVA or VF.
- **CONCLUSIONS:** Subacute deterioration of BCVA and/or VF following acute NAION is not uncommon while optic disc edema is present, with sizeable proportion of patients experiencing dramatic visual decline. Deterioration in visual function within the first 10 weeks of presentation does not exclude the diagnosis of NAION and further investigations should only be performed if additional clinical features are discordant with this diagnosis. (Am J Ophthalmol 2024;268: 258–262. © 2024 Else-

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INTRODUCTION

NONARTERITIC ANTERIOR ISCHEMIC OPTIC NEUROPATHY (NAION) is the most common cause of optic neuropathy in patients over 50 years of age. Typical presentation involves sudden onset painless vision loss, often altitudinal, with clinical findings of a swollen optic disc and contralateral “disc at risk”. Unfortunately, there is no known treatment for NAION and the visual loss can be devastating.

Without any serologic or radiographic markers available to confirm NAION, the diagnosis can sometimes be brought into question, especially in cases of severe vision loss or subacute worsening of vision after initial presentation.¹ In these situations, concern may be raised for other treatable diagnoses requiring targeted and timely therapy, such as arteritic anterior ischemic optic neuropathy, optic neuritis, or compressive optic neuropathy. The fear of missing these diagnoses can prompt clinicians to initiate extensive, invasive, and costly workups, including blood tests, magnetic resonance imaging (MRI) of the orbits with contrast, and temporal artery biopsies. It can also lead to unnecessary treatment with very high doses of corticosteroids, which may have very serious side effects.

The Ischemic Optic Neuropathy Decompression Trial (IONDT) showed that vision progressively deteriorates in almost one-third of patients with NAION within the first 30 days after symptom onset.² In contrast, Hayreh and Zimmerman found worsening of BCVA in only 4% of eyes presenting with vision $\leq 20/70$ and in 9% of eyes with $\geq 20/60$ vision between presentation and resolution of disc edema.³ There were several smaller studies conducted at the end of the 20th century that examined aspects of progressive visual loss in NAION eyes, but there are no other studies evaluating both the degree of BCVA and visual field (VF) progression within 10 weeks after initial presentation of NAION. This is the time period during which disc edema typically resolves.⁴ Knowing the frequency and severity of further

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visual loss during this time is essential for making appropriate clinical decisions regarding workup and management.

The purpose of this study was to determine what percentage of eyes with acute NAION demonstrate worsening of BCVA and peripheral VF in the first 10 weeks following diagnosis, and the degree to which this deterioration occurs.

METHODS

This was a retrospective descriptive chart review of all patients at an academic neuro-ophthalmology practice who were diagnosed with acute NAION between February 2014 and December 2023. Institutional Review Board (IRB)/Ethics Committee approval was obtained from the University of Toronto for a retrospective study. The study adhered to the tenets of the Declaration of Helsinki. Inclusion criteria were: (1) diagnosis of NAION made within four weeks of symptom onset, (2) documentation of disc edema at presentation, (3) disc at risk in the fellow eye or negative investigations for other causes of optic neuropathy, (4) follow up visit within 10 weeks of initial assessment documenting resolution of disc edema, and (5) no other identified cause for disc edema. All patients in the appropriate age group had normal inflammatory markers. As previous literature has shown that optic disc edema in NAION resolves between 5.8 to 11.4 weeks after onset of vision loss, 10 weeks from initial presentation was selected as the follow-up limit for this study.⁴ Patients were excluded if they were missing essential clinical data or had any confounding ocular disease that could impact BCVA or VF in the affected eye.

The electronic medical records (EMR) were searched for the diagnosis of “NAION” and all identified charts were reviewed to determine eligibility. Data collected for eligible patients included demographics (age at presentation and sex at birth), days from symptom onset to presentation, affected eye(s), character of vision loss (blurry vision, shadow in vision, or complete loss of vision), course of vision loss prior to assessment (subjective worsening, improvement, or stability of vision), physical examination findings (BCVA on Snellen chart, laterality of disc edema, disc at risk in fellow eye), results of diagnostic testing (optical coherence tomography (OCT) average retinal nerve fiber layer (RNFL) thickness, VF mean deviation (MD), type of VF defect in affected eye), follow-up data (weeks from initial presentation to follow-up, BCVA, average RNFL thickness, and MD in affected eye), and additional testing performed in patients without disc-at-risk in the fellow eye.

Data analysis was descriptive and performed using Microsoft Excel Statistics. Continuous variables were reported as mean or median values with a range from minimum to maximum or standard deviation (SD). Categorical variables were described as percentages. For each affected eye, it was determined if BCVA worsened by ≥ 2 Snellen lines,

TABLE 1. Snellen Visual Acuity Measurements Used to Determine Number of Lines Lost or Gained When Assessing Best Corrected Visual Acuity at Follow-Up

Snellen Visual Acuity
20/20
20/25
20/30
20/40
20/50
20/60
20/80
20/100
20/125
20/160
20/200
20/400
20/800
Counting fingers
Hand motions

*Adapted from Hosal et al. (2000)⁶ and Junn et al. (2022).^{4,5}

≥ 4 Snellen lines, or ≥ 8 Snellen lines at follow-up. A VA chart adapted from Junn et al. and Hosal et al. was used to determine number of lines for change in BCVA (Table 1).^{5,6} A threshold change in BCVA of ≥ 2 Snellen lines was selected based on previous studies and its clinical significance regarding quality of life.^{7,8} Data from each eye was also analyzed to determine if MD worsened by ≥ 2 decibels (dB) or ≥ 4 dB. This threshold was selected because previous glaucoma studies found that in stable patients with end-stage VF defects, MD remained within 1.3 dB with a 99% confidence interval, and therefore worsening of ≥ 2 db should reflect true progression.⁹ Finally, descriptive changes in VF patterns were analyzed in eyes where MD declined past threshold.

Further analysis was done to determine how many patients experienced either worsening of BCVA or MD, worsening of BCVA but stable MD, worsening of MD but stable BCVA, or worsening of both simultaneously. For those patients whose VA worsened by ≥ 8 Snellen lines, the charts were further reviewed to determine if they underwent workup beyond checking inflammatory markers (C-reactive protein, erythrocyte sedimentation rate, and platelets).

RESULTS

After searching EMR, 205 patients were identified of which 46 met inclusion and exclusion criteria. The majority of patients were excluded due to presenting more than 4 weeks after symptom onset or being seen in follow-up more than 10 weeks after initial assessment. Two patients experienced

TABLE 2. Additional Workup Beyond Inflammatory Markers Performed in Patients With BCVA Decline of ≥ 8 Snellen Lines

Age	Sex	Symptom Onset to Initial Exam (Days)	Initial BCVA	Time to Follow-Up (Weeks)	Follow-Up BCVA	Imaging	Blood Work	Other
51	M	5	20/40	10	CF	Not done	Not done	N/A
57	M	11	20/20	2	CF	MRI orbits with gadolinium normal	Anti-NMO & -MOG antibodies negative	N/A
62	F	4	20/25	4	20/400	MRI orbits with contrast, MRI spine, CT body noncontributory	ANA, VDRL, ACE, anti-NMO & -MOG antibodies all normal	Temporal artery biopsy negative

sequential NAION and one experienced simultaneous bilateral NAION, therefore a total of 49 eyes were included. Sixty-seven percent of patients were male and average age at presentation was 66 years (range 39-88 years). Mean time from symptom onset to presentation was 13 days ($SD \pm 7$ days), and median time was 12 days. In 57% NAION occurred in the right eye. Forty-eight percent complained of a shadow in their vision, 33% blurry vision, 6% complete loss of vision, and the remainder had nonspecific visual complaints. Sixty-one percent reported stability of vision loss prior to assessment, 35% reported worsening, and 4% reported improvement.

Mean BCVA in the affected eye at presentation was between 20/70 to 20/80 Snellen acuity (range 20/20 to hand motions). All patients had optic disc edema in the affected eye, with mean RNFL thickness of 198 μm ($SD \pm 55 \mu m$). Ninety percent of affected eyes had a disc at risk in the fellow eye, and the remaining 10% ($n = 5$) underwent MRI orbits with contrast and serum testing for antemyelin oligodendrocyte (MOG) and antineuromyelitis optica (NMO) antibodies to rule out other causes of optic neuropathy, which were normal in all patients. All patients in the appropriate age group had inflammatory markers checked to rule out giant cell arteritis. Seventy-six percent of patients demonstrated an altitudinal or arcuate VF defect, 16% demonstrated global depression, and the remainder demonstrated central or nonspecific loss. Average mean deviation at presentation was -14.07 dB ($SD \pm 8.38$ dB). All patients were seen in follow-up within 10 weeks of initial presentation (mean 5.3 weeks, $SD \pm 2.6$ weeks).

Mean BCVA at follow-up was between 20/80 and 20/100 Snellen acuity (range 20/20 to hand motions). Twenty-two percent of eyes ($n = 11$) demonstrated worsening of BCVA by ≥ 2 Snellen lines. Of these eyes, 55% ($n = 6$) worsened by ≥ 4 lines and 27% ($n = 3$) worsened by ≥ 8 lines; mean loss of vision was 5 lines. Most eyes, (67%, $n = 33$) demonstrated stable BCVA and in 10% ($n = 5$) vision improved by ≥ 2 Snellen lines. Two of the three patients whose BCVA declined by ≥ 8 Snellen lines underwent a workup beyond inflammatory markers (Table 2).

Average MD on VF testing at follow-up was -14.6 dB. Twenty-seven percent ($n = 13$) of eyes demonstrated worsening of MD by ≥ 2 dB and 18% ($n = 9$) by ≥ 4 dB; mean worsening was -6.4 dB ($SD \pm 3.75$ dB). Regarding change in type of VF defect, 46% ($n = 6$) developed increased density of initial field loss, 31% ($n = 4$) evolved from an altitudinal/arcuate defect to global depression involving central vision, and 23% ($n = 3$) developed a new altitudinal/arcuate defect affecting previously uninvolved portion of VF. Corresponding change in segmental disc edema was not observed in these patients. In total, 41% ($n = 20$) of eyes demonstrated either worsening of BCVA by ≥ 2 lines or worsening of MD by ≥ 2 dB. Table 3 illustrates the breakdown of patients with decline in BCVA and/or VF.

DISCUSSION

This study provides clinicians with information about the expected proportion of eyes with acute NAION that will experience clinically significant worsening of VA or VF within the first 10 weeks after presentation. While dramatic worsening in acuity and fields can be very disconcerting for both the patient and clinician, our results indicate that this is a common occurrence in NAION and is not inconsistent with this diagnosis. Almost a quarter (22%) of patients with acute NAION had worsening of BCVA ≥ 2 Snellen lines within 10 weeks of presentation. In over 50% of these patients the vision worsened by ≥ 4 lines, and in one third of patients it worsened by ≥ 8 lines. Analysis of VF demonstrated that nearly one third of patients had worsening of VF defect by ≥ 2 dB of MD, and in 20% of these patients the defect worsened by ≥ 4 dB. In total, 41% of eyes demonstrated a clinically significant decline in either VA or VF. However, only 20% of eyes with decline in either BCVA or VF demonstrated decline in both, underscoring the importance of monitoring both acuity and fields in these patients. While the majority of VF progression was due to increased density of the original defect, involvement

TABLE 3. Demonstration of Eyes With Visual Function Stability Versus Decline in Regard to Visual Acuity and Visual Fields (Percent of Total Eyes in Study Shown in Round Brackets [$n = 49$], and Percent of Eyes With Any Visual Decline Shown in Square Brackets [$n = 20$])

	BCVA Stable	BCVA Decline ≥ 2 Snellen Lines	Total Eyes
VF MD stable	29 (59%)	7 (14%) (35%)	36 (73%)
VF MD decline ≥ 2 dB	9 (18%) (45%)	4 (8%) (20%)	13 (27%) (65%)
Total eyes	38 (78%)	11 (22%) (55%)	49 (100%)

of the previously uninvolved altitudinal field or progression to global depression with loss of central vision also occurred. Together, these findings illustrate how visual function can decline within 10 weeks of diagnosis of NAION while disc edema might still be present.

Several studies at the end of the 20th century looked at the short-term natural history of vision loss in eyes with NAION. Results from these studies found a decline of BCVA or VF in 8%-22% of eyes in the 24 hours to 30 days after initial presentation.^{7,10,11} This range is in accordance with our study's results, however, none of the previous studies delineated the proportion of eyes that deteriorated by varying degrees of acuity or field loss. Borchert and Lessell reported a case series of 10 patients with worsening of VA and/or VF following NAION, several of which experienced dramatic worsening within a short interval from onset, similar to a proportion of eyes with visual decline identified in our study.¹²

The IONDT found that 30% of patients with better than 20/64 BCVA at initial assessment showed progression below 20/64 up to 30 days later. These results were limited by inclusion of patients with only modestly reduced BCVA ($>20/64$) at presentation and the degree of visual loss was not quantified.² Hayreh and Zimmerman conducted a prospective study on the natural history of NAION looking at visual outcomes from presentation to time of optic disc edema resolution.³ Of those assessed within two weeks of symptom onset, 4% of eyes with presenting BCVA of $\leq 20/70$ and 9% of eyes with presenting BCVA $\geq 20/60$ worsened by ≥ 3 Snellen lines; 14% with moderate to severe VF loss and 23% with minimal to mild loss worsened by ≥ 0.5 dB. The proportion of eyes with BCVA decline in Hayreh's study is much less than that in our study, possibly due to a higher threshold (≥ 3 versus ≥ 2 Snellen lines) for visual loss that they used, however, their reported proportion of patients with VF decline was very similar to our results despite their much lower threshold for VF change (≥ 0.5 versus ≥ 2 dB).³ None of the aforementioned studies delineated degree of worsening in both acuity and fields, as was done in the study reported here.

It is thought that ischemia in NAION occurs at the level of the short posterior ciliary arteries supplying the optic disc.¹⁴ A triggering event, such as nocturnal hypotension,

may result in poor blood flow through these small vessels causing ischemic damage to the optic nerve axons, axoplasmic flow stasis, and resultant swelling of the optic nerve head (ONH).¹⁵ A small cup-to-disc ratio makes the nerve especially susceptible to this scenario. We hypothesize that visual decline after initial presentation occurred in 41% of patients because while the ONH remains swollen, further compression, ischemia and axoplasmic flow stasis of the remaining ganglion cell axons in the optic disc can occur in a cyclical fashion. Unfortunately, despite numerous treatments being studied for NAION, including high dose corticosteroid therapy, antiplatelets and anticoagulants, as well as surgical optic nerve decompression, none have proven effective.^{13,16-20}

The findings of this study are important in guiding physicians diagnosing patients with suspected NAION. Investigations such as MRI orbits with gadolinium, serological tests, and temporal artery biopsy should not be performed in patients with suspected NAION only because they demonstrate subacute worsening of BCVA or VF. In this study, two of the three patients with a decline in BCVA ≥ 8 Snellen lines had extensive negative workup to rule out alternate causes of optic neuropathy (Table 2).

Limitations of this study include its retrospective nature. As a consequence, in order to maintain strict inclusion criteria, many patients who did not meet said criteria had to be excluded limiting the sample size. Worsening of acuity may have been missed in those patients who presented longer after symptom onset (35% of patients reported subjective decline in visual function prior to being assessed). Another significant limitation is that patients seen in neuro-ophthalmology practice likely represent more complex presentations of NAION than those seen by comprehensive ophthalmologists, and this may skew the results. However, the demographic characteristics of patients with NAION in this study were very similar to those previously reported by the IONDT (mean age at onset 66 years in both studies, 72% male in ONDT and 67% in this study),² suggesting that the population studied here is reflective of typical NAION. Further prospective studies with larger numbers would be helpful to identify potential features which increase the risk and degree of subacute visual decline in eyes with NAION.

CONCLUSIONS

NAION is the most common optic neuropathy in patients over age 50 years without any biomarkers or clinically proven treatment. Deterioration of visual acuity or peripheral visual fields following initial presentation is not uncommon and cumulatively was seen in 41% of patients in this study. Severe deterioration of BCVA ≥ 4 Snellen lines was seen in 12% of eyes and severe progression of MD ≥ 4 dB in 18% of eyes. Therefore, progressive decline in visual function within the first 10 weeks following initial assessment does not preclude the diagnosis of NAION and further investigations should only be performed if additional clinical features are discordant with this diagnosis.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

HEATHER M. MCDONALD: Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **MOIZ LAKHANI:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **ANGELA KWAN:** Writing – original draft, Resources, Investigation, Formal analysis, Data curation. **ARMIN HANDZIC:** Writing – original draft, Investigation, Formal analysis, Conceptualization. **EDWARD MARGOLIN:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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